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CIS 4374

Homework 1

The Initial Kickoff for Project Documentation (Version 1.0)

Vision And Scope:

Project Acquisition- Our company \$oftware Corp. Wants to branch out into different avenues of software/app development that can help improve another aspect of people's needs in our modern world. The route we identified that best fits this project campaign is to improve how parking in urban areas/cities are and manage them using a digital platform that helps provide parking information in these areas in real-time. This is a problem we want to solve with our app and make it a smart parking platform that can be universal across the nation and make it easier for users to have at their disposal, for now we plan to launch in the Houston area. There is parking systems that are in place already but often tend to have proprietary bottlenecks that lack efficient tools for monitoring parking occupancy and managing parking reservations as well as different payment systems that make frustrating for parking users. We aim to improve the parking experience for users and operators of parking areas by having an efficient system that benefits all parties involved.

Vision Statement- \$oftware Corps new smart parking platform promises a convenient method for users to locate, reserve, and pay for parking spaces on our app or website which in turn will reduce the time that is spent looking for parking and further improve parking facility utilization in urban/city areas. It will also be convenient for parking operators overseeing the parking areas/garages user park their vehicles by using the smart parking platform system to keep payments, vehicles parked and spaces available/unavailable logged.

Project Scope, this project will include the following:

- User account registration and login
- Real-time parking availability
- Parking space search

- Parking reservations
- Navigation assistance
- Digital payment processing
- Alerts and notifications
- Occupancy reporting and analytics
- Administrative dashboard
- Support line for user/manager
- AI helper tool

We will not be focusing on this:

- Maintenance services for vehicles
- Physical construction of parking garages (we are a digital service, if parking garages want to use our services we will have a contracts area to be contacted via website)

Competitive analysis research:

This is a competitive market, there is an abundance of many parking services and management systems in place. Here is some competitors that already have prominence in this market.

ParkMobile- This is the most popular app for parking services that is widely used where users can access app or website and have real-time parking sessions. I'd say there's little to no errors for this app.

Spot Hero- This is another popular app for parking services that specializes in advanced reservations with a large network of parking garages in big city's and also shows price comparisons of each garage. Availability varies by city though so not available in every city.

PayByPhone- This parking app is similar to the ones mentioned prior, it works similar to ParkMobile where a user can find parking using their app or website and have real time parking sessions, but it's not available in every state and you have to manually type in vehicle info and code.

What can Software Corp do to differentiate amongst competitors:

- Use AI-based parking recommendations like an assistant within the app or website to help user
- Have dynamic pricing suggestions to show which place to park has the best pricing
- Real-time occupancy tracking that you see in modern parking garages but within the app or website
- Have a unified website and mobile experience where users can navigate easily
- Have a predictive availability forecasting for available parking, especially for events like concerts or sports games

Initial Stakeholders List:

- Drivers and vehicle owners
 - ^Interested in finding parking in a quick and convenient manner
- Parking facility operators
 - ^Interested in increasing utilization of their parking facilities
- City transportation departments
 - ^Interested in reducing congestion
- System administrators
 - ^Interested in the maintenance of the system
- Finance and billing personnel
 - ^Interested in the processing of the payment
- Mobile application users
 - ^Interested in a convenient app form factor that makes finding and paying for parking easier
- Third-party payment providers
 - ^Interested in providing different methods of payment to users while getting monetary gain from service

Initial Use Cases:

Use case 1: Downloading the app or searching thru web browser

The Actor: User/Driver

The Goal: finding the application or website

Structure of flow:

1. Enter the name of the application that Software Corp will use (sometimes can be found by entering developer name too) on Appstore or web browser.
2. Once found, download to phone or enter site
3. You are now on the platform

Use case 2: Registering for an account

The actor: User/Driver

The Goal: Creating an account for parking app

Structure of flow:

1. As user enter your information into platform
2. The platform's system will proceed to validate information inputted
3. The user's account is created

Use case 3: Finding parking

The Actor: User/Driver

The Goal: Find available parking at desired location

Structure of flow:

1. The user/driver enters desired destination
2. The platform's system will then display nearby parking garages
3. User/driver views availability in parking garages

Use case 4: Reserving the parking spot

The Actor: User/Driver

The Goal: User/Driver will reserve their spot on app

Use Case 5: Making the payment

The Actor: User/Driver

The Goal: The user/driver will make the payment for however long they are parking for on the app

Use Case 6: Using the apps AI assistant

The Actor: User/Driver

The Goal: Being able to find and use AI assistant on the app

SRS Functional/Non-Functional Requirements

Functional requirements:

- We want to allow users to successfully register and log in securely
- The system will be able to display parking availability in real-time
- The system has access to data reports
- The system allows reservation for parking ahead of time
- The system must be able to process any form of online payments via app/website

Non-Functional Requirements:

- The system must support up to 20,000 concurrent users
- The system has to provide 99.99% uptime

- The system must update any and all inventory values in 1 second
- The safety firewall security must block out any unknown sources of payment

References:

Parkmobile parking app: Find & Pay For Parking: On-Street & Reservation. ParkMobile. (n.d.).
<https://parkmobile.io/>

Get everywhere easier: Reserve parking and save. SpotHero. (n.d.-a). <https://spothero.com/>

Simplifying your journey. PayByPhone. (n.d.). <https://www.paybyphone.com/>

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Homework 2

Deliverables (WBS/Draft Timeline) (Version 2.0)

Work Breakdown Structure (WBS)

1.0 User Access Management

1.1 New User Registration

- • 1.1.1 A Sign-Up Interface for our new users
- • 1.1.2 Information Verification to make sure our users are real and not bots
- • 1.1.3 Profile Setup to help make the user have an easier method of logging on

1.2 User Authentication

- • 1.2.1 Login Portal that helps secure username and password
- • 1.2.2 Identity Confirmation to help confirm if its the original users account
- • 1.2.3 User Session Control that will aid in logging out a user if account stays idle

2.0 Parking Discovery and Booking

2.1 Parking Spot Search

- • 2.1.1 Area Lookup to find areas to park
- • 2.1.2 Interactive Map View that users can see where they can find specific parking spots
- • 2.1.3 Space Availability Check that will help in the time management aspect of being efficient for the user

2.2 Space Reservation

- • 2.2.1 Reserve Parking Space that a user can do beforehand when going to an event in downtown
- • 2.2.2 Update Reservation Details incase there is a change in user or vehicle used
- • 2.2.3 Remove Reservation if user does not wish to continue with reservation

3.0 Transaction Management

3.1 Online Payments

- • 3.1.1 Payment Service Configuration to help apply users preferred method of payment
- • 3.1.2 Transaction Authorization that will act as a safeguard for fraud
- • 3.1.3 Digital Receipt Delivery that will help get a digital receipt to the user

3.2 Payment Tracking

- • 3.2.1 Transaction Records to show what a user paid for
- • 3.2.2 Financial Summaries that summarize the total time parked and how much was paid

4.0 Management Console

4.1 Operations Administration

- • 4.1.1 Parking Facility Oversight to make sure everyone parks in legitimate manner
- • 4.1.2 Capacity Monitoring to make sure there is no overflow of vehicles
- • 4.1.3 Customer Account Administration to make sure there is no fraud in fake customer alias

4.2 Reporting Tools

- • 4.2.1 Utilization Reports to help report facility, customer, parking reports, etc.
- • 4.2.2 Income Analysis Reports to show how much a parking garage or area is making depending on the owners
- • 4.2.3 System Activity Metrics to keep metrics organized and in order

5.0 Alerts and Route Assistance

5.1 User Alerts

- • 5.1.1 Booking Confirmation Notices to alert users
- • 5.1.2 Payment Status Alerts to help alert users of things regarding to payment
- • 5.1.3 Parking Schedule Reminders to remind users of how long they have for parking

5.2 Route Assistance

- • 5.2.1 GPS Service Connection to help guide users to parking locations

- 5.2.2 Turn-by-Turn Directions to help get the most efficient way of getting to a parking area
- 5.2.3 Arrival Support so that there can be a system of help and guidance for users that park

Project Schedule

Task ID	Task	Duration
1	Requirements Gathering	1 Week
2	Solution Architecture Design	1 Week
3	User Authentication Development	1 Week
4	Parking Search Module Development	1 Week
5	Reservation Feature Development	1 Week
6	Payment System Integration	1 Week
7	Management Dashboard Development	1 Week
8	Notification & Navigation Features	1 Week
9	System Verification & Testing	1 Week
10	Project Closure Review	1 Week

Dependencies

- Solution Architecture Design will start after the Requirements Gathering.
- User Authentication Development will start after the Solution Architecture Design.
- Parking Search Module Development will start after the User Authentication Development.
- Reservation Feature Development will start after Parking Search Module Development.
- Payment System Integration will start after the Reservation Feature Development.
- Management Dashboard Development will start after the Payment System Integration.
- Notification & Navigation Features will start after the Dashboard Development.
- System Verification & Testing will start once all development work is finished.
- Project Closure Review will then start after System Verification & Testing.

Gantt Chart (Draft)

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
Requirements Analysis	X	X								
System Design		X	X							
Authentication Development			X	X						
Parking Search Development			X	X	X					
Reservation System Development				X	X	X				
Payment Integration					X	X				
Administrative Dashboard						X	X			
Notifications & Navigation						X	X	X		
Testing & Quality Assurance							X	X	X	
Final Project Review										X

Updated Dependencies

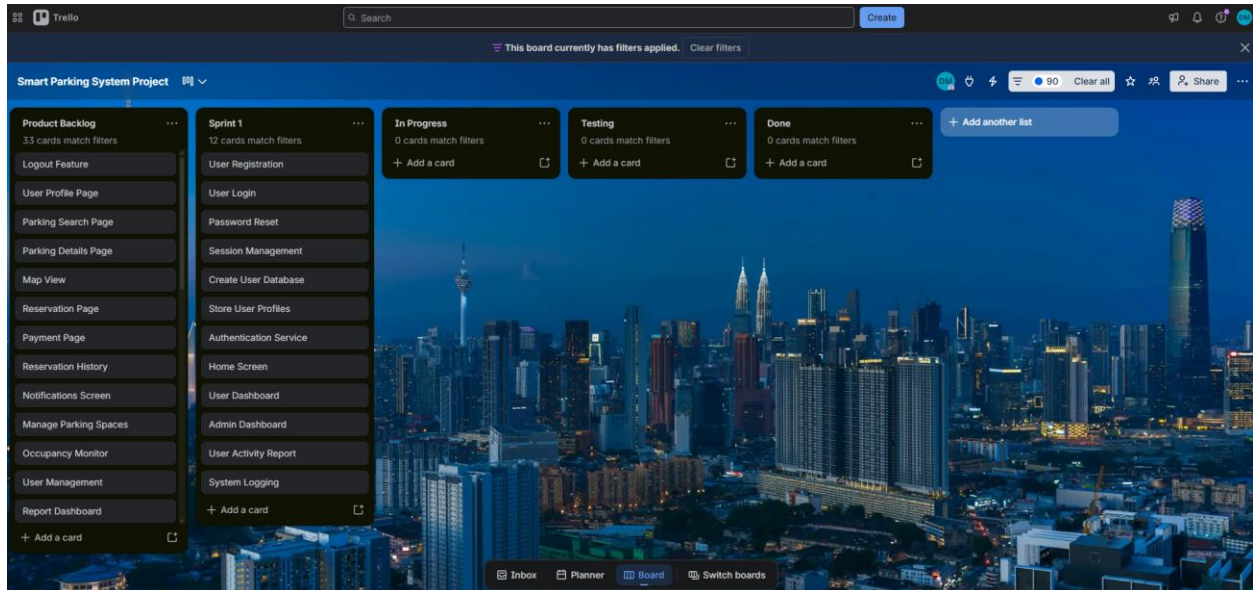
- System design starts during the latter part of the requirements analysis.
- Authentication development will start once the core design decisions are completed.
- Parking Search Development will start running in parallel with Authentication Development.
- Reservation System Development begins while Parking Search Development will still underway.
- Payment integration will start after the reservation functionality is stable.

- The Administrative Dashboard and Notifications/Navigation portion will be developed concurrently with later integration work.
- Testing begins before all development is completely finished to allow early defect identification and make sure the app is fully functional.
- Final Project Review will then commence after Testing & QA portion is completed.

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Homework 3

Product Backlog (Trello)



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Homework 4

Risk Register and communication plan

Part 1: Risk Identification

Technical Risks:

- 1.The GPS integration may not be able to provide accurate directions at all times; errors are bound to happen at some point.
- 2.The payment gateway may be prone to fail during transactions.
- 3.The system could experience downtime due to any server issues that may occur.
- 4.The user authentication could have potential security vulnerabilities that can put a users data at risk.

Schedule Risks:

- 1.There can be delays with developing the reservation system.
- 2.Integration of the payment services could take longer than expected due to many factors in infrastructure.
- 3.The testing phase could uncover major defects and bugs in software/hardware
- 4.The requirement changes can potentially cause timeline adjustments that may need to adapt too.

Financial Risks:

1. The payment gateway service fees could potentially exceed budget if funds go over.
- 2.There will be additional costs involving cloud hosting.
- 3.There can potentially be unexpected software licensing expenses involved.
- 4.There can be potential overruns in the budget due to delays in development.

People Risks:

- 1.Team members could be unavailable depending on different circumstances.

2. There can be a lack of experience when it comes to GPS API's.
3. There can be potential issues with communication between the stakeholders and the developers.
4. The possibility of a workload imbalance among the team members can occur.

Part 2: Risk Register

ID	Risk Description	Probability	Impact	Owner	Response Strategy	Status
R1	GPS navigation inaccuracies	Medium	High	The Development Team	To test with multiple mapping services	Open
R2	Payment integration failure	Medium	High	The Backend Developer	Use sandbox testing before deployment so its functional	Open
R3	Server downtime	Low	High	The System Administrator	To configure the backup servers	Open
R4	Authentication security issues	Medium	High	The Security Lead	To implement the encryption and security testing	Open
R5	Reservation module delays	Medium	Medium	The Project Manager	To monitor the progress weekly	Open
R6	Extended testing period	Medium	Medium	The QA Team	To begin testing earlier in the project	Open

R7	Budget overrun	Low	High	The Project Manager	To track the costs throughout the project	Open
R8	Unexpected hosting expenses	Medium	Medium	The Project Manager	To maintain the budget reserve	Open
R9	Team member absence	Medium	Medium	Team Lead	To cross-train the team members	Open
R10	Poor stakeholder communication	Medium	Medium	Project Manager	To schedule the regular status meetings	Open

Part 3: Communication Plan

Communication Type	Audience	Frequency	Method	Responsible Party
Daily Team Check-In	Development Team	Daily	Microsoft Teams / Zoom	Team Lead
Sprint Planning Meeting	Project Team	Bi-Weekly	Teams Meeting	Project Manager
Progress Status Report	Stakeholders	Weekly	Email	Project Manager
Risk Review Meeting	Project Team	Weekly	Teams Meeting	Project Manager
Testing Review Meeting	QA & Developers	Weekly During Testing	Teams Meeting	QA Lead

Final Project Review	Stakeholders	End of Project	Presentation	Project Manager
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